

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc. I.T Semester V)
Applied Business Analytics
Practical – VI

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6. Decision Making under Uncertainty:

a. Develop a decision tree model to make business decisions considering uncertainties and associated probabilities at each decision point.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("supermarket_sales.csv")
```

```
# Q1. Display first 10 records
print("\nQ1. First 10 Records:")
print(df.head(10))
```

```
# Create revenue column using Total
df["revenue"] = df["Total"]
```

```
# Q2. Dataset information
print("\nQ2. Dataset Information:")
print("Number of Records:", df.shape[0])
print("Number of Columns:", df.shape[1])
print("Column Names:")
print(df.columns.tolist())
print("\nData Types:")
print(df.dtypes)
```

```
# Q3. Unique product lines
print("\nQ3. Unique Product Lines:")
print(df["Product line"].unique())
```

```
# Q4. Total revenue by product line
total_revenue = df.groupby("Product line")["revenue"].sum()
print("\nQ4. Total Revenue by Product Line:")
print(total_revenue)
```

```
# Q5. Average revenue by product line
```

```
# Q5. Average revenue by product line
average_revenue = df.groupby("Product line")["revenue"].mean()
print("\nQ5. Average Revenue by Product Line:")
print(average_revenue)
```

```
# Q6. Total quantity sold by product line
total_quantity = df.groupby("Product line")["Quantity"].sum()
print("\nQ6. Total Quantity Sold:")
print(total_quantity)
```

```
# Q7. Median revenue
median_revenue = df["revenue"].median()
print("\nQ7. Median Revenue:", median_revenue)
```

```
# Q8. Classify transactions
df["revenue_level"] = df["revenue"].apply(
    lambda x: "High Revenue" if x >= median_revenue else "Low Revenue"
)
```

```
# Q9. Display revenue_level
print("\nQ9. Revenue Level:")
print(df[["revenue", "revenue_level"]].head(10))
```

```
# Q10. Number of High/Low transactions
count_table = pd.crosstab(df["Product line"], df["revenue_level"])
print("\nQ10. High and Low Revenue Transactions:")
print(count_table)
```

```
# Q11. Probability of High/Low Revenue
probability_table = count_table.div(count_table.sum(axis=1), axis=0)
print("\nQ11. Revenue Probabilities:")
print(probability_table)
```

```
# Q12. Verify probabilities add to 1
probability_sum = probability_table.sum(axis=1)
print("\nQ12. Probability Sum:")
print(probability_sum)

# Q13. Average revenue for High/Low Revenue
payoff_table = df.groupby(
    ["Product Line", "revenue_level"]
)[ "revenue"].mean().unstack()

print("\nQ13. Average Revenue / Payoff:")
print(payoff_table)

# Q14. Create DataFrame with Product Line, Revenue Level,
# Probability and Payoff
decision_data = []

for product in probability_table.index:
    for level in probability_table.columns:
        decision_data.append([
            product,
            level,
            probability_table.loc[product, level],
            payoff_table.loc[product, level]
        ])

decision_df = pd.DataFrame(
    decision_data,
    columns=["Product Line", "Revenue Level", "Probability", "Payoff"]
)

print("\nQ14. Decision DataFrame:")
print(decision_df)
```

```
# Q15. Decision alternatives and uncertain outcomes
print("\nQ15. Decision Alternatives:")
print(probability_table.index.tolist())

print("\nUncertain Outcomes:")
print(probability_table.columns.tolist())

# Q16 & Q17. Decision Tree Data
print("\nQ16-Q17. Decision Tree:")
for product in probability_table.index:
    print("\nDecision:", product)
    for level in probability_table.columns:
        print(
            "    ->", level,
            "| Probability:", round(probability_table.loc[product, level], 4),
            "| Payoff:", round(payoff_table.loc[product, level], 2)
        )

# Q18. Total Revenue Visualization
plt.figure(figsize=(10, 6))
total_revenue.sort_values().plot(kind="barh")
plt.title("Total Revenue by Product Line")
plt.xlabel("Total Revenue")
plt.ylabel("Product Line")
plt.tight_layout()
plt.show()
```

```
# Q19. Probability Visualization
probability_table.plot(
    kind="bar",
    figsize=(10, 6)
)
plt.title("Probability of High and Low Revenue")
plt.xlabel("Product Line")
plt.ylabel("Probability")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Q20. Highest probability of High Revenue
highest_high_probability = probability_table["High Revenue"].idxmax()
highest_probability = probability_table["High Revenue"].max()

print("\nQ20. Highest Probability of High Revenue:")
print(highest_high_probability, "=", round(highest_probability, 4))
```

Output:

Q1. First 10 Records:

	Invoice ID	Branch	City	Customer type	Gender	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3081	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	
5	699-14-3026	C	Naypyitaw	Normal	Male	
6	355-53-5943	A	Yangon	Member	Female	
7	315-22-5665	C	Naypyitaw	Normal	Female	
8	665-32-9167	A	Yangon	Member	Female	
9	692-92-5582	B	Mandalay	Member	Female	

	Product line	Unit price	Quantity	Tax 5%	Total	Date	\
0	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	
1	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019	
2	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	
3	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	
4	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	
5	Electronic accessories	85.39	7	29.8865	627.6165	3/25/2019	
6	Electronic accessories	68.84	6	20.6520	433.6920	2/25/2019	
7	Home and lifestyle	73.56	10	36.7800	772.3800	2/24/2019	
8	Health and beauty	36.26	2	3.6260	76.1460	1/10/2019	
9	Food and beverages	54.84	3	8.2260	172.7460	2/20/2019	

Q2. Dataset Information:

Number of Records: 1000

Number of Columns: 18

Column Names:

['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender', 'Product line',

Data Types:

Invoice ID	str
Branch	str
City	str
Customer type	str
Gender	str
Product line	str
Unit price	float64
Quantity	int64
Tax 5%	float64
Total	float64
Date	str
Time	str
Payment	str
cogs	float64
gross margin percentage	float64
gross income	float64
Rating	float64
revenue	float64

dtype: object

Q3. Unique Product Lines:

<StringArray>

```
[ 'Health and beauty', 'Electronic accessories', 'Home and lifestyle',
  'Sports and travel', 'Food and beverages', 'Fashion accessories']
```

Length: 6, dtype: str

Q4. Total Revenue by Product Line:

Product line

```
Electronic accessories    54337.5315
Fashion accessories       54305.8950
Food and beverages       56144.8440
Health and beauty        49193.7390
Home and lifestyle       53861.9130
Sports and travel        55122.8265
```

Name: revenue, dtype: float64

Q5. Average Revenue by Product Line:

Product line

```
Electronic accessories    319.632538
Fashion accessories       305.089298
Food and beverages       322.671517
Health and beauty        323.643020
Home and lifestyle       336.636956
Sports and travel        332.065220
```

Name: revenue, dtype: float64

Q10. High and Low Revenue Transactions:

revenue_level	High Revenue	Low Revenue
---------------	--------------	-------------

Product line

Electronic accessories	79	91
Fashion accessories	84	94
Food and beverages	86	88
Health and beauty	81	71
Home and lifestyle	82	78
Sports and travel	88	78

Q11. Revenue Probabilities:

revenue_level	High Revenue	Low Revenue
---------------	--------------	-------------

Product line

Electronic accessories	0.464706	0.535294
Fashion accessories	0.471910	0.528090
Food and beverages	0.494253	0.505747
Health and beauty	0.532895	0.467105
Home and lifestyle	0.512500	0.487500
Sports and travel	0.530120	0.469880

Q14. Decision DataFrame:

	Product Line	Revenue Level	Probability	Payoff
0	Electronic accessories	High Revenue	0.464706	537.006152
1	Electronic accessories	Low Revenue	0.535294	130.923577
2	Fashion accessories	High Revenue	0.471910	512.329375
3	Fashion accessories	Low Revenue	0.528090	119.896037
4	Food and beverages	High Revenue	0.494253	515.127680
5	Food and beverages	Low Revenue	0.505747	134.589358
6	Health and beauty	High Revenue	0.532895	497.221407
7	Health and beauty	Low Revenue	0.467105	125.616972
8	Home and lifestyle	High Revenue	0.512500	529.582738
9	Home and lifestyle	Low Revenue	0.487500	133.796519
10	Sports and travel	High Revenue	0.530120	518.546438
11	Sports and travel	Low Revenue	0.469880	121.676154

Q15. Decision Alternatives:

['Electronic accessories', 'Fashion accessories', 'Food and beverages',

Q6. Total Quantity Sold:

Product line

```
Electronic accessories    971
Fashion accessories       902
Food and beverages       952
Health and beauty        854
Home and lifestyle       911
Sports and travel        920
```

Name: Quantity, dtype: int64

Q7. Median Revenue: 253.848

Q9. Revenue Level:

revenue revenue_level

```
0  548.9715  High Revenue
1   80.2200  Low Revenue
2  340.5255  High Revenue
3  489.0480  High Revenue
4  634.3785  High Revenue
5  627.6165  High Revenue
6  433.6920  High Revenue
7  772.3800  High Revenue
8   76.1460  Low Revenue
```

Q12. Probability Sum:

Product line

```
Electronic accessories    1.0
Fashion accessories       1.0
Food and beverages       1.0
Health and beauty        1.0
Home and lifestyle       1.0
Sports and travel        1.0
```

dtype: float64

Q13. Average Revenue / Payoff:

revenue_level High Revenue Low Revenue

Product line

```
Electronic accessories    537.006152  130.923577
Fashion accessories       512.329375  119.896037
Food and beverages       515.127680  134.589358
Health and beauty        497.221407  125.616972
Home and lifestyle       529.582738  133.796519
Sports and travel        518.546438  121.676154
```

Uncertain Outcomes:
['High Revenue', 'Low Revenue']

Q16-Q17. Decision Tree:

Decision: Electronic accessories

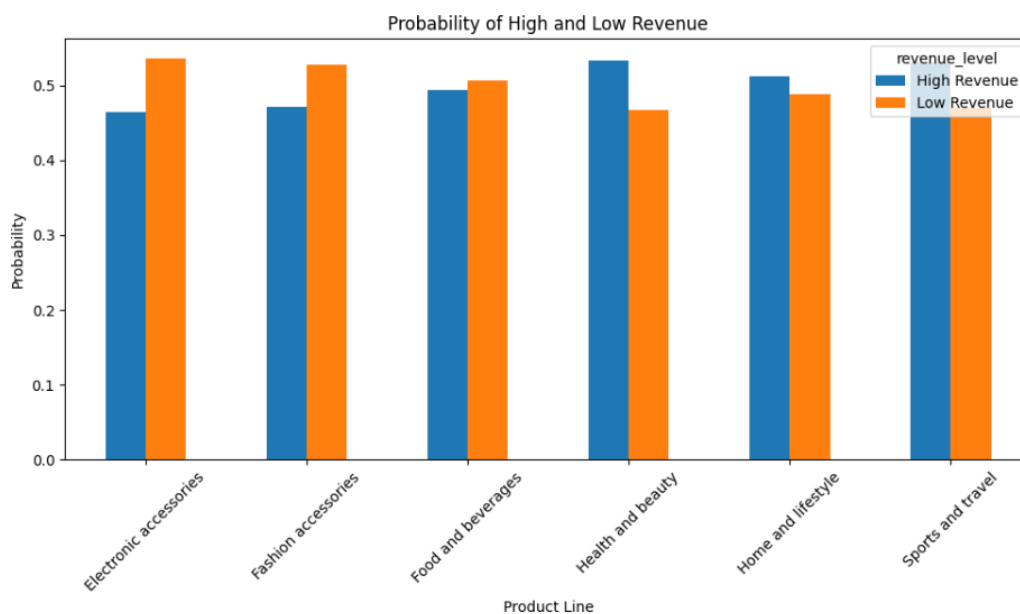
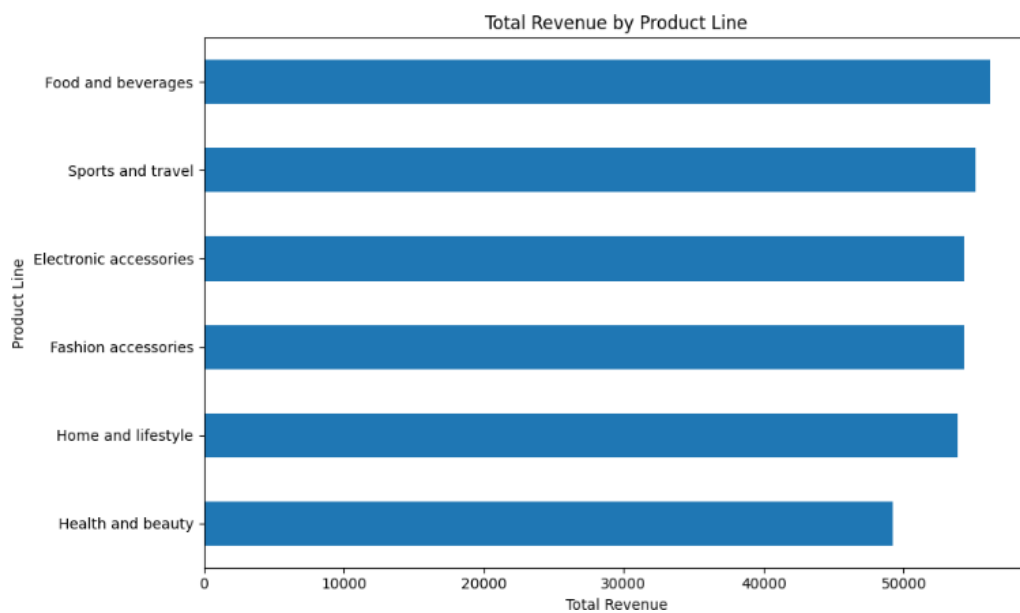
-> High Revenue | Probability: 0.4647 | Payoff: 537.01
-> Low Revenue | Probability: 0.5353 | Payoff: 130.92

Decision: Fashion accessories

-> High Revenue | Probability: 0.4719 | Payoff: 512.33
-> Low Revenue | Probability: 0.5281 | Payoff: 119.9

Decision: Food and beverages

-> High Revenue | Probability: 0.4943 | Payoff: 515.13
-> Low Revenue | Probability: 0.5057 | Payoff: 134.59



Q20. Highest Probability of High Revenue:
Health and beauty = 0.5329

b. Apply the concept of expected value to evaluate different decision alternatives and select the optimal one.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("supermarket_sales.csv")

# Use Total as Revenue
df["revenue"] = df["Total"]

# Calculate median revenue
median_revenue = df["revenue"].median()

# Classify revenue
df["revenue_level"] = df["revenue"].apply(
    lambda x: "High Revenue" if x >= median_revenue else "Low Revenue"
)

# Calculate probabilities
count_table = pd.crosstab(df["Product line"], df["revenue_level"])
probability_table = count_table.div(count_table.sum(axis=1), axis=0)

# Calculate payoffs
payoff_table = df.groupby(
    ["Product line", "revenue_level"]
)[["revenue"]].mean().unstack()

# PART B: APPLY EXPECTED VALUE

# Q21. Explain Expected Value
print("\nQ21. Expected Value:")
print("Expected Value is used to compare different product lines.")
print("The product line with the highest Expected Value is selected.")
```

```
# Q22. Formula
print("\nQ22. Formula:")
print("EV = (P(High Revenue) * High Revenue Payoff) +")
print("      (P(Low Revenue) * Low Revenue Payoff)")

# Q23. Calculate Expected Value of one product line
product = probability_table.index[0]

high_probability = probability_table.loc[product, "High Revenue"]
low_probability = probability_table.loc[product, "Low Revenue"]

high_payoff = payoff_table.loc[product, "High Revenue"]
low_payoff = payoff_table.loc[product, "Low Revenue"]

ev_one = (high_probability * high_payoff) + \
         (low_probability * low_payoff)

print("\nQ23. Expected Value of", product, ":")
print(round(ev_one, 2))

# Q24. Calculate Expected Value for every product line
expected_values = (
    probability_table["High Revenue"] * payoff_table["High Revenue"]
    + probability_table["Low Revenue"] * payoff_table["Low Revenue"]
)

print("\nQ24. Expected Value for Every Product Line:")
print(expected_values.round(2))
```

```
# Q25. Create Expected Value DataFrame
ev_df = pd.DataFrame({
    "Product Line": probability_table.index,
    "High Revenue Probability":
        probability_table["High Revenue"].values,
    "High Revenue Payoff":
        payoff_table["High Revenue"].values,
    "Low Revenue Probability":
        probability_table["Low Revenue"].values,
    "Low Revenue Payoff":
        payoff_table["Low Revenue"].values,
    "Expected Value":
        expected_values.values
})

print("\nQ25. Expected Value DataFrame:")
print(ev_df.round(2))

# Q26. Compare Expected Values
print("\nQ26. Comparison of Expected Values:")
print(
    ev_df[["Product Line", "Expected Value"]]
    .sort_values("Expected Value", ascending=False)
    .round(2)
)
```

```
# Q27. Highest Expected Value
highest_ev = ev_df.loc[ev_df["Expected Value"].idxmax()]

print("\nQ27. Highest Expected Value:")
print("Product Line:", highest_ev["Product Line"])
print("Expected Value:", round(highest_ev["Expected Value"], 2))

# Q28. Lowest Expected Value
lowest_ev = ev_df.loc[ev_df["Expected Value"].idxmin()]

print("\nQ28. Lowest Expected Value:")
print("Product Line:", lowest_ev["Product Line"])
print("Expected Value:", round(lowest_ev["Expected Value"], 2))

# Q29. Rank Product Lines
ranking = ev_df.sort_values(
    "Expected Value",
    ascending=False
).reset_index(drop=True)

ranking["Rank"] = ranking.index + 1

print("\nQ29. Ranking:")
print(
    ranking[
        ["Rank", "Product Line", "Expected Value"]
    ].round(2)
)
```

```
# Q30. Bar Chart of Expected Values
plt.figure(figsize=(10, 6))
plt.bar(
    ev_df["Product Line"],
    ev_df["Expected Value"]
)
plt.title("Expected Value of Each Product Line")
plt.xlabel("Product Line")
plt.ylabel("Expected Value")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Q31. Compare highest High Revenue probability
highest_probability_product = probability_table[
    "High Revenue"
].idxmax()

highest_ev_product = expected_values.idxmax()

print("\nQ31. Comparison:")
print(
    "Highest probability of High Revenue:",
    highest_probability_product
)
print(
    "Highest Expected Value:",
    highest_ev_product
)

# Q32. Are they the same?
print("\nQ32. Result:")
```

```
print(
    "New High Revenue Probability:",
    round(new_high_probability, 4)
)
print(
    "New Low Revenue Probability:",
    round(new_low_probability, 4)
)
print(
    "New Expected Value:",
    round(new_ev_probability, 2)
)

# Q34. Change payoff and recalculate EV
old_high_payoff = payoff_table.loc[
    product, "High Revenue"
]

new_high_payoff = old_high_payoff + 50

new_ev_payoff = (
    probability_table.loc[product, "High Revenue"]
    * new_high_payoff
    +
    probability_table.loc[product, "Low Revenue"]
    * payoff_table.loc[product, "Low Revenue"]
)

print("\nQ34. Changed Payoff:")
print("Product Line:", product)
print(
    "New High Revenue Payoff:",
    round(new_high_payoff, 2)
)
```

```
if highest_probability_product == highest_ev_product:
    print(
        "Yes, both are the same product line."
    )
else:
    print(
        "No, they are different product lines."
    )

# Q33. Change probability and recalculate EV
product = highest_ev_product

old_high_probability = probability_table.loc[
    product, "High Revenue"
]

new_high_probability = old_high_probability + 0.10

if new_high_probability > 1:
    new_high_probability = 1

new_low_probability = 1 - new_high_probability

new_ev_probability = (
    new_high_probability *
    payoff_table.loc[product, "High Revenue"]
    +
    new_low_probability *
    payoff_table.loc[product, "Low Revenue"]
)

print("\nQ33. Changed Probability:")
print("Product Line:", product)
```

```
print(
    "New Expected Value:",
    round(new_ev_payoff, 2)
)

# Q35. Automatically select product with highest EV
best_product = expected_values.idxmax()
best_ev = expected_values.max()

print("\nQ35. Automatic Decision:")
print("Selected Product Line:", best_product)
print("Highest Expected Value:", round(best_ev, 2))

# Q36. Final Decision Tree with Expected Value
print("\nQ36. Final Decision Tree:")

for product in probability_table.index:

    print("\n", product)

    print(
        "└─ High Revenue | Probability:",
        round(
            probability_table.loc[
                product, "High Revenue"
            ], 4
        ),
        "| Payoff:",
        round(
            payoff_table.loc[
                product, "High Revenue"
            ], 2
        )
    )
```

```

    ), 2
)

print(
    "    Low Revenue | Probability:",
    round(
        probability_table.loc[
            product, "Low Revenue"
        ], 4
    ),
    "| Payoff:",
    round(
        payoff_table.loc[
            product, "Low Revenue"
        ], 2
    ),
)

print(
    "    Expected Value:",
    round(expected_values.loc[product], 2)
)

# Q37. Business Recommendation
print("\nQ37. Business Recommendation:")

print(
    "The recommended product line is",
    best_product,
    "because it has the highest Expected Value of",
    round(best_ev, 2),
    "."
)

```

✓ 0.2s

Output:

Q21. Expected Value:
Expected value is used to compare different product lines.
The product line with the highest Expected Value is selected.

Q22. Formula:

$$EV = (P(\text{High Revenue}) \times \text{High Revenue Payoff}) + (P(\text{Low Revenue}) \times \text{Low Revenue Payoff})$$

Q23. Expected Value of Electronic accessories :
319.63

Q24. Expected Value for Every Product Line:
Product line
 Electronic accessories 319.63
 Fashion accessories 305.09
 Food and beverages 322.67
 Health and beauty 323.64
 Home and lifestyle 336.64
 Sports and travel 332.07
 dtype: float64

Q25. Expected Value DataFrame:

	Product Line	High Revenue Probability	High Revenue Payoff
0	Electronic accessories	0.46	537.01
1	Fashion accessories	0.47	512.33
2	Food and beverages	0.49	515.13
3	Health and beauty	0.53	497.22
4	Home and lifestyle	0.51	529.58
5	Sports and travel	0.53	518.55

Q26. Comparison of Expected Values:

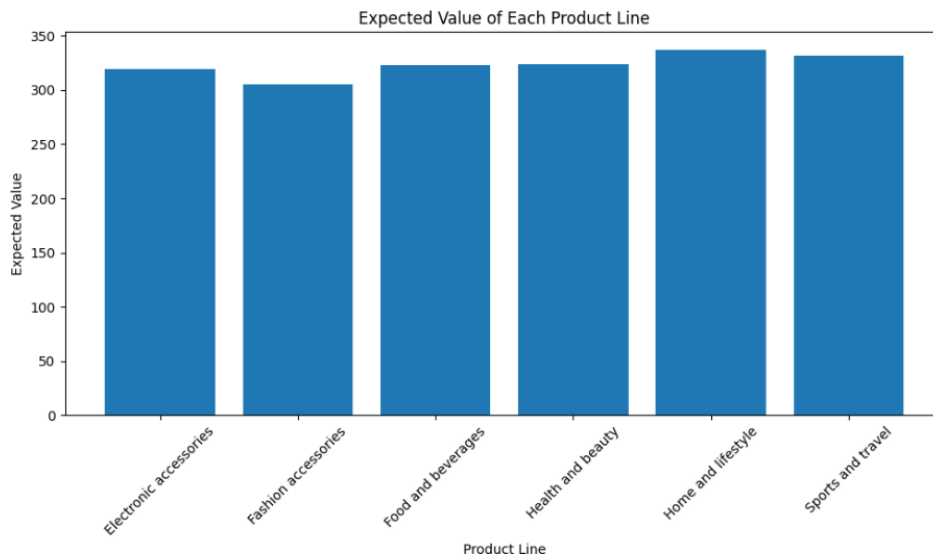
	Product Line	Expected Value
4	Home and lifestyle	336.64
5	Sports and travel	332.07
3	Health and beauty	323.64
2	Food and beverages	322.67
0	Electronic accessories	319.63
1	Fashion accessories	305.09

Q27. Highest Expected Value:
Product Line: Home and lifestyle
Expected Value: 336.64

Q28. Lowest Expected Value:
Product Line: Fashion accessories
Expected Value: 305.09

Q29. Ranking:

	Rank	Product Line	Expected Value
0	1	Home and lifestyle	336.64
1	2	Sports and travel	332.07
2	3	Health and beauty	323.64
3	4	Food and beverages	322.67
4	5	Electronic accessories	319.63
5	6	Fashion accessories	305.09



Q31. Comparison:

Highest probability of High Revenue: Health and beauty
 Highest Expected Value: Home and lifestyle

Q32. Result:

No, they are different product lines.

Q33. Changed Probability:

Product Line: Home and lifestyle
 New High Revenue Probability: 0.6125
 New Low Revenue Probability: 0.3875
 New Expected Value: 376.22

Q34. Changed Payoff:

Product Line: Home and lifestyle
 New High Revenue Payoff: 579.58
 New Expected Value: 362.26

Q35. Automatic Decision:

Selected Product Line: Home and lifestyle
 Highest Expected Value: 336.64

Q36. Final Decision Tree:

Electronic accessories

└─ High Revenue	Probability: 0.4647	Payoff: 537.01
└─ Low Revenue	Probability: 0.5353	Payoff: 130.92
Expected Value: 319.63		

Home and lifestyle

└─ High Revenue	Probability: 0.5125	Payoff: 529.58
└─ Low Revenue	Probability: 0.4875	Payoff: 133.8
Expected Value: 336.64		

Sports and travel

└─ High Revenue	Probability: 0.5301	Payoff: 518.55
└─ Low Revenue	Probability: 0.4699	Payoff: 121.68
Expected Value: 332.07		

Q37. Business Recommendation:

The recommended product line is Home and lifestyle because it has the highest Expected Value of 336.64 .

Conclusion:

The Decision Making under Uncertainty practical successfully analyzed the supermarket sales dataset using decision tree concepts and Expected Value. The revenue of each product line was classified into High Revenue and Low Revenue based on the median revenue, and the probabilities and payoffs of the outcomes were calculated. Expected Value was then used to compare all product lines. The analysis showed that Home and lifestyle has the highest Expected Value and is therefore the best product line to select for the business decision. This practical demonstrates how probability, payoff, and Expected Value can be used to make effective business decisions under uncertainty.